1. Experiment in short

Number of nodes	5
Number of topologies	4
Location of the experiment	Garda Lake, Italy $(45.50413°\text{N}, 10.7233°\text{E})$
Duration of each measurement	951 seconds
Link sampling interval	1 second
Link quality measurements	RSSI

2. Location

The experiment was conducted in Dec. 2015, in the Garda lake, located in northern Italy. A Google map illustration of the experimental setup is shown in Fig. 1.

The experiment included 5 nodes, communicating using EvoLogics modems. The nodes were deployed from harbor docks and from boats. By moving the two boats, we created the four different network topologies shown in Fig. 1. Nodes 2 and 3 were deployed from harbor docks 75 m from each other and at water depth of 2 m and 4 m, respectively (all topologies). Nodes 1 and 5 were placed 10 m from each other and were deployed from a small boat at water depth 10 m and distance $\sim$700 m from node 2 (Topologies 1, 2, 3) and at water depth 2 m and distance $\sim$50 m from node 2 (Topology 4). Node 4 was deployed from an additional boat at water depth 5 m and distance 200 m (Topology 1), 700 m (Topology 2), and 600 m (Topologies 3, 4) from node 2. When the boats were distant from the harbor, the water depth was roughly 30 m, whereas the water depth at the harbor was between 2 and 4 m.

3. Topologies

4. Description of the nodes

The modems were EvoLogics SC2R 18/34 units, operating in the 18–34 kHz frequency band using EvoLogics's proprietary sweep-spread carrier modulation scheme. For further information, the interested reader is referred to EvoLogics's web site: https://www.evologics.de/en/products/acoustics/index.html

5. Tested protocols and schemes

The deployment was employed to test a special TDMA schedule that exploited the knowledge of near-far node pairs in underwater acoustic networks. This resulted in additional transmission opportunities for each node, and thus in a denser transmission scheduling, and in a more frequent sampling of the link quality over time.

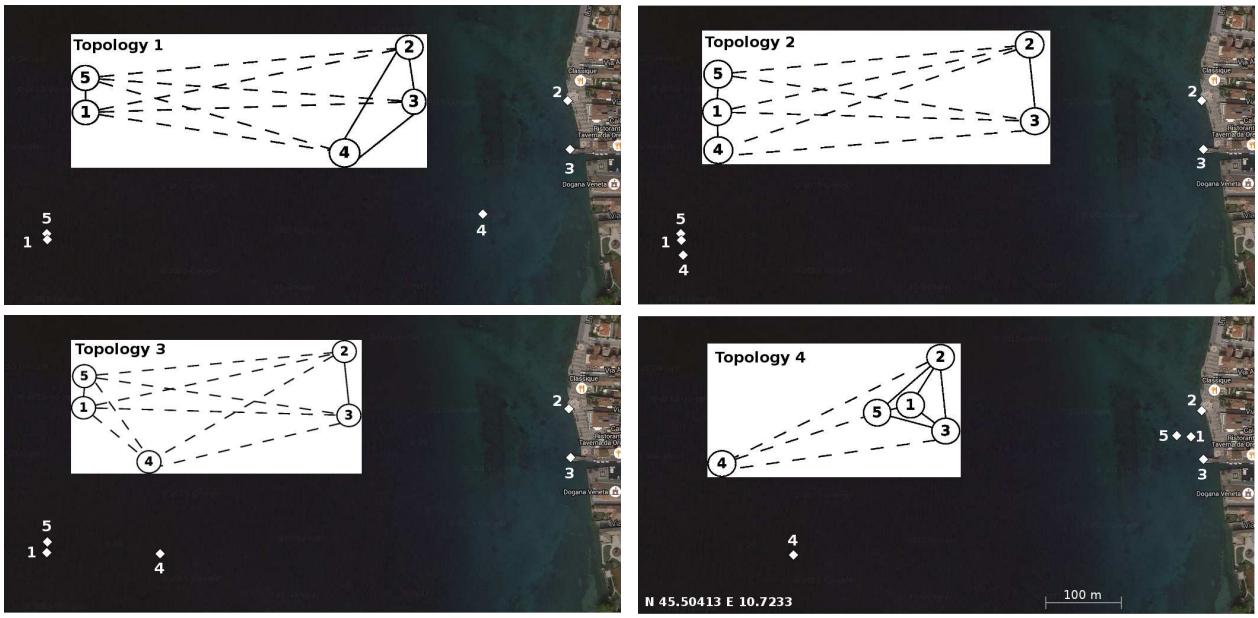


Fig. 1. Setting of the Garda lake experiment. The white panel in each map conveys the logical topology of the network. The bottom-left corner in all maps corresponds to the geographical coordinates $(45.50413°\text{N}, 10.7233°\text{E})$. (Maps courtesy of Google Maps.)

6. Link quality measurements

The network links are evaluated by means of the received signal strength indicator (RSSI). Transmissions were carried out using the EvoLogics modems' instant message mode, which allows the user to send data while excluding any further control exerted by the modem's DMAC protocol. Upon transmission, the RSSI was returned directly by the receiving modem's firmware.

7. Available data

The measured data is provided in file `GardaLake-12-15.mat`, which includes all topologies presented in Fig. 1. For each topology, we provide 951 measurement epochs, which have been taken at constant intervals of 1 second starting from the beginning of each experiment. The file contains four types of matrices:

- the location matrices `LocMat1`...`LocMat4`, of size $951\times5\times3$;
- the topology measurement matrices `TopMat1`...`TopMat4`, of size $951\times5\times5$;
- the adjacency matrices `AdjMat1`...`AdjMat4` of size $951\times5\times5$;
- the technology matrices `TechMat1`...`TechMat4` of size $951\times5\times1$.

Additionally, for each matrix type, we concatenate all matrices into a single matrix, respectively named:

- `FullLocMat` of size $3804\times5\times3$;
- `FullTopMat` of size $3804\times5\times5$;

- `FullAdjMat` of size $3804 \times 5 \times 5$;
- `FullTechMat` of size $3804 \times 5 \times 1$.

Using Matlab/Octave matrix notation:

- `LocMat1(t,i,:)`...`LocMat4(t,i,:)` or `FullLocMat(t,i,:)` are the UTM coordinates (zone 32T) and the depth of node i at time instant t;
- `TopMat1(t,i,j)`...`TopMat4(t,i,j)` or `FullTopMat(t,i,j)` is the RSSI for link $i \to j$ at time t;
- `AdjMat1(t,i,j)`...`AdjMat4(t,i,j)` or `FullAdjMat(t,i,j)` is equal to 1 if there exists a link between nodes i and j at time t;
- `TechMat1(t,i,p)`...`TechMat4(t,i,p)` or `FullTech(t,i,p)` is equal to 1 if node i can use physical layer technology p at time t.

8. Reference to detailed description

Further details about this experiment are given in the following paper:

- R. Diamant, P. Casari, F. Campagnaro and M. Zorzi, "Leveraging the NearFar Effect for Improved Spatial-Reuse Scheduling in Underwater Acoustic Networks," in *IEEE Transactions on Wireless Communications*, vol. 16, no. 3, pp. 1480–1493, March 2017. DOI: 10.1109/TWC.2016.2646682